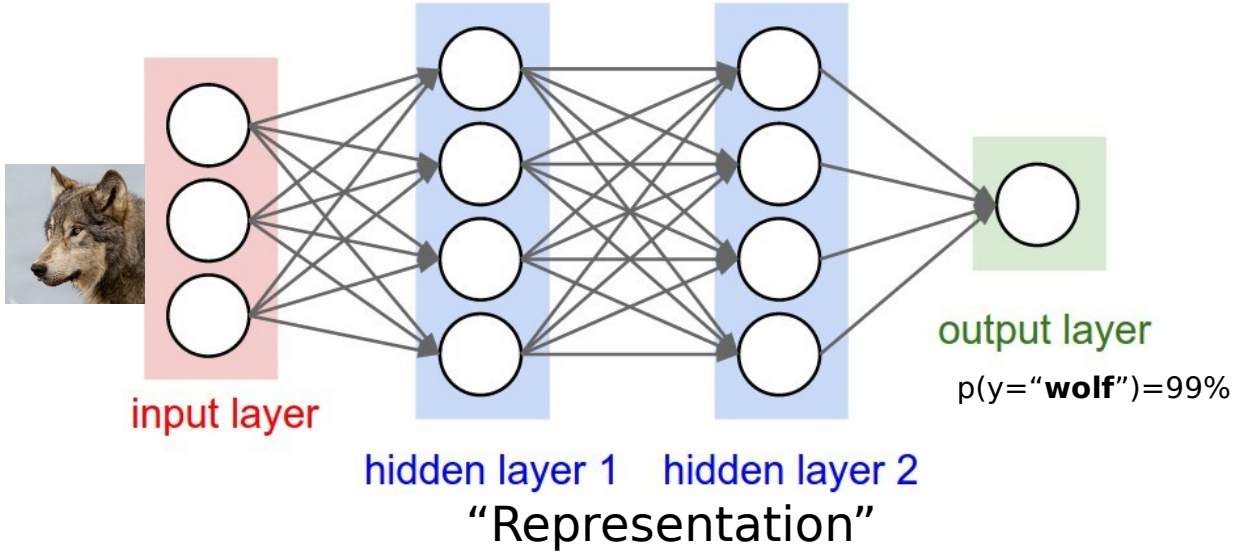
CS 229
Problems with vision and uncertainty in ML



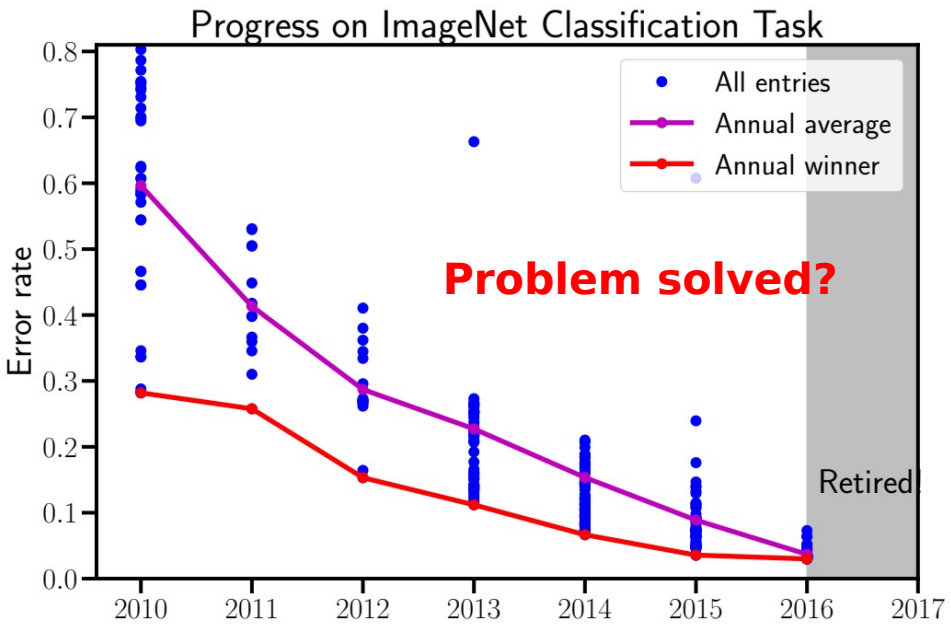
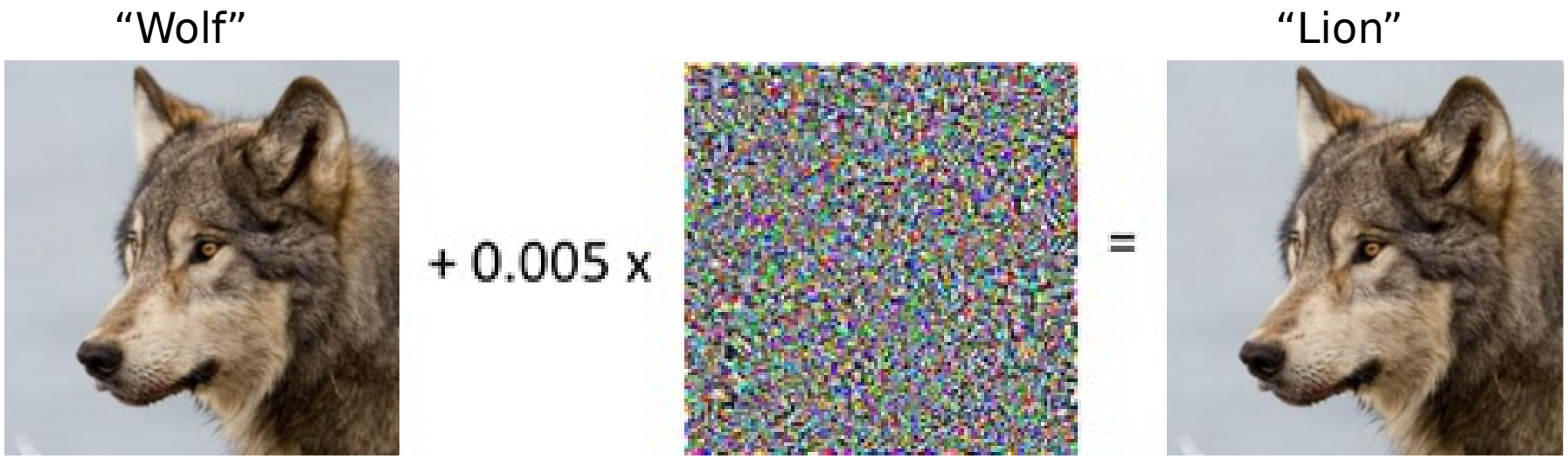


Image classification benchmarks were easy and artificial

Information used by neural nets and humans is not the same!

Hard problems remain.

Sensitivity to adversarial perturbations

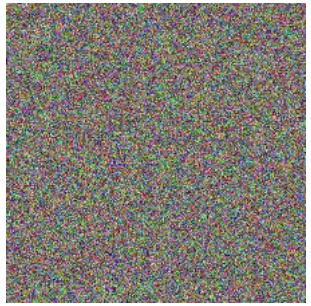


Real world attacks

https://arxiv.org/pdf/1707.08945.pdf

Another issue – far Out Of Distribution (OOD) inputs

p(y=“**lion**”)=99%



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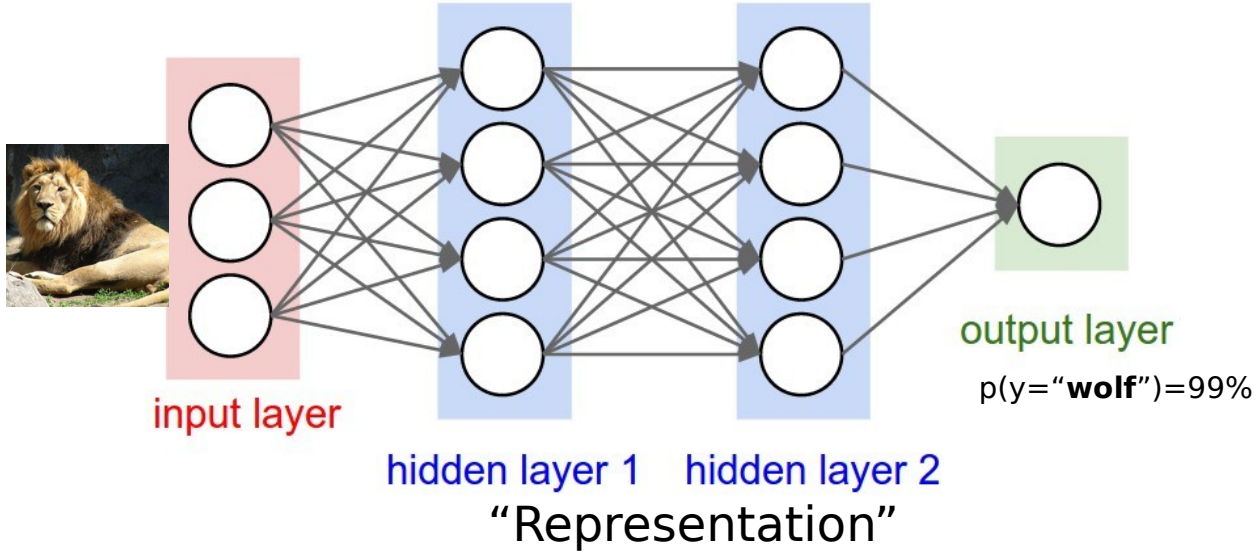
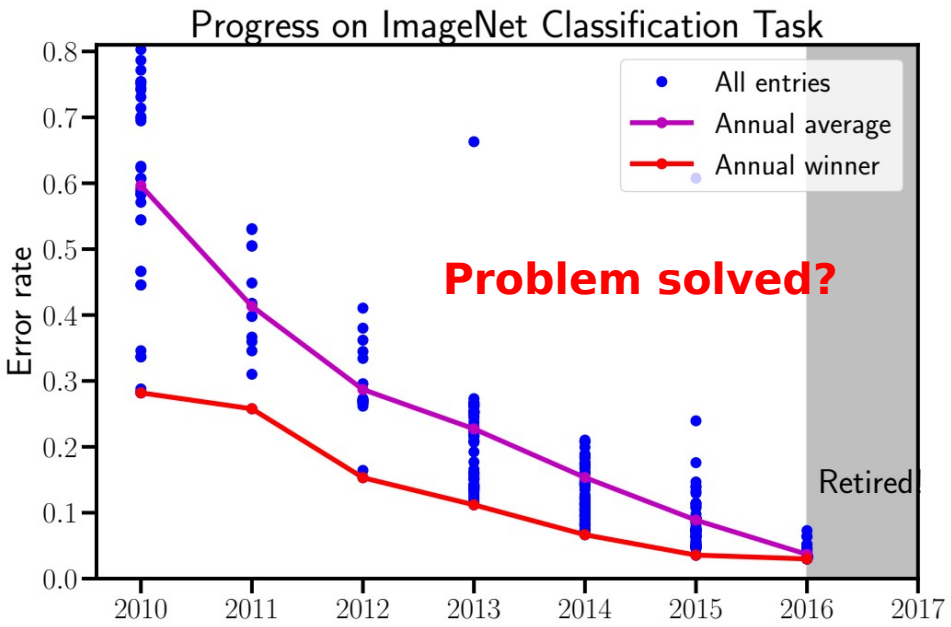
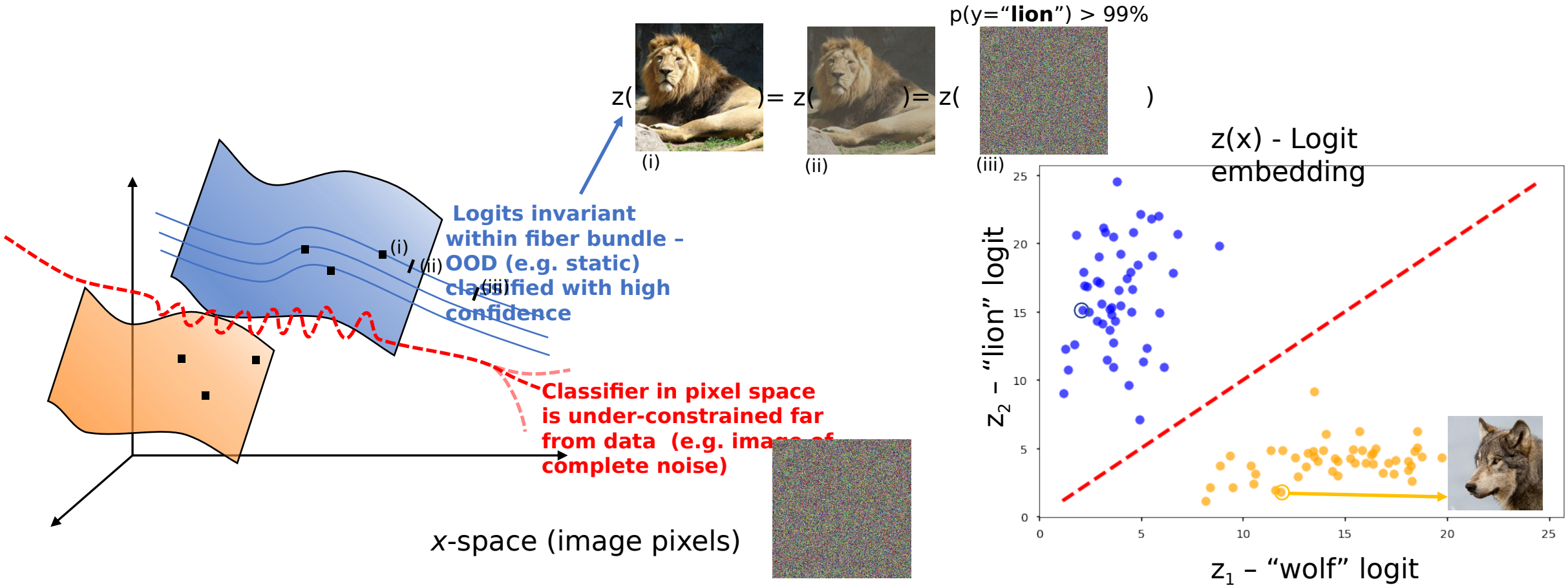
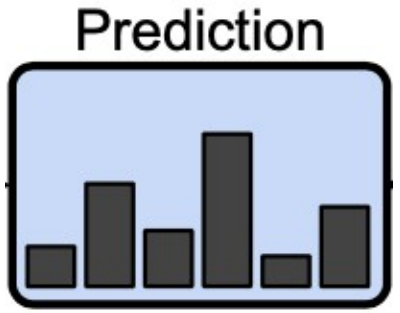


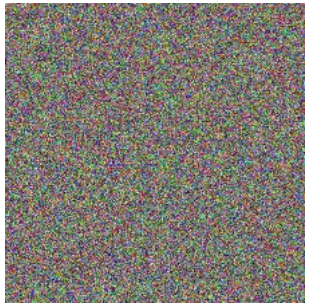



p(y="lion") > 99%
z( )= z( )= z( )
(i)
(ii)
(iii)
Logits invariant within fiber bundle – OOD (e.g. static) classified with high confidence
Classifier in pixel space is under-constrained far from data (e.g. image of complete noise)
x-space (image pixels)
z(x) - Logit embedding
z_2 – "lion" logit
z_1 – "wolf" logit
0
5
10
15
20
25

What do the probabilities that our neural nets give us really *mean*?



p(y=“**lion**”)=99%



p(y=“**lion**”)=99%

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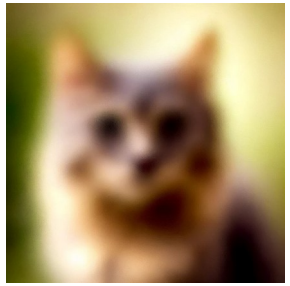


p(y=“**cat”**)=80%

We often say that the probability a neural net assigns to the top choice is its **“confidence”.** If a neural net is 80% confident of a prediction, will that prediction be correct 80% of the time?

Confidence calibration game: give a range that you think the answer lies in with probability 50%

Martin Luther King’s age at death:

Length of the Nile River:

Number of countries that are members of OPEC:

Number of books in the Old Testament:

My guesses

Martin Luther King’s age at death:

35-55

Length of the Nile River:

700-1500 miles

Number of countries that are members of OPEC:

9-23

Number of books in the Old Testament:

30-55

Correct answers

39

6737 km or 4187 m

12 (as of 2026)

39

Primer on Uncertainty & Robustness

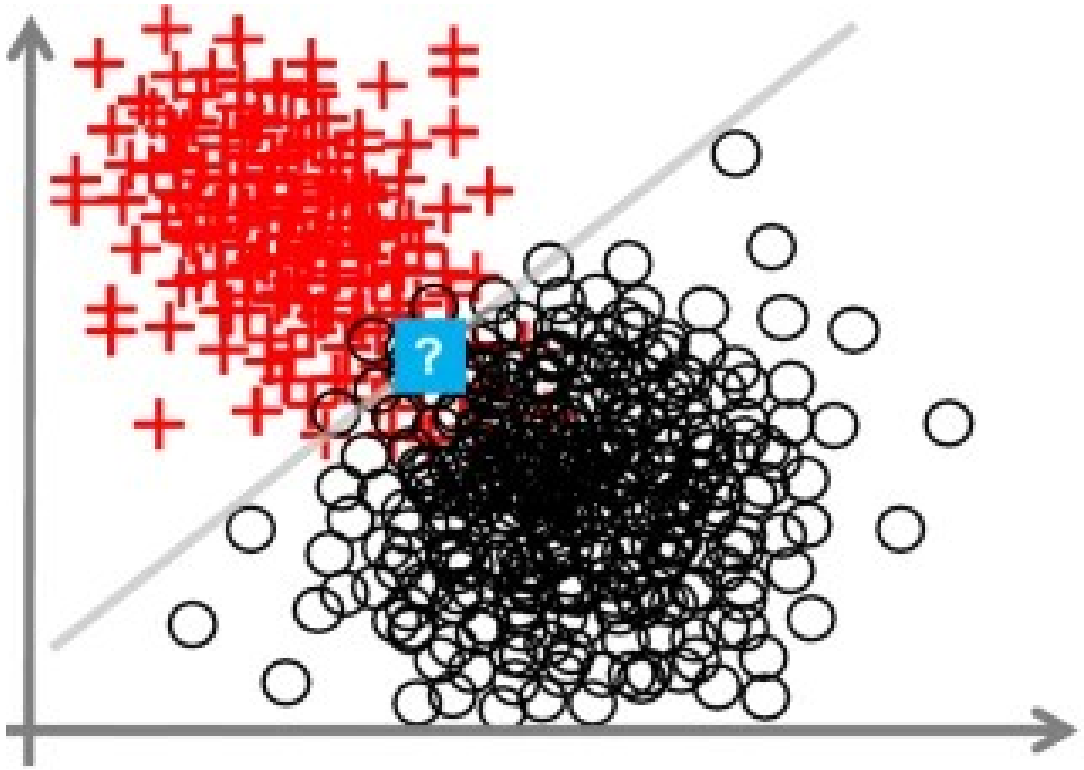
Many of my slides on this are from
Balaji Lakshminarayanan

You can tell which ones because of the logo:

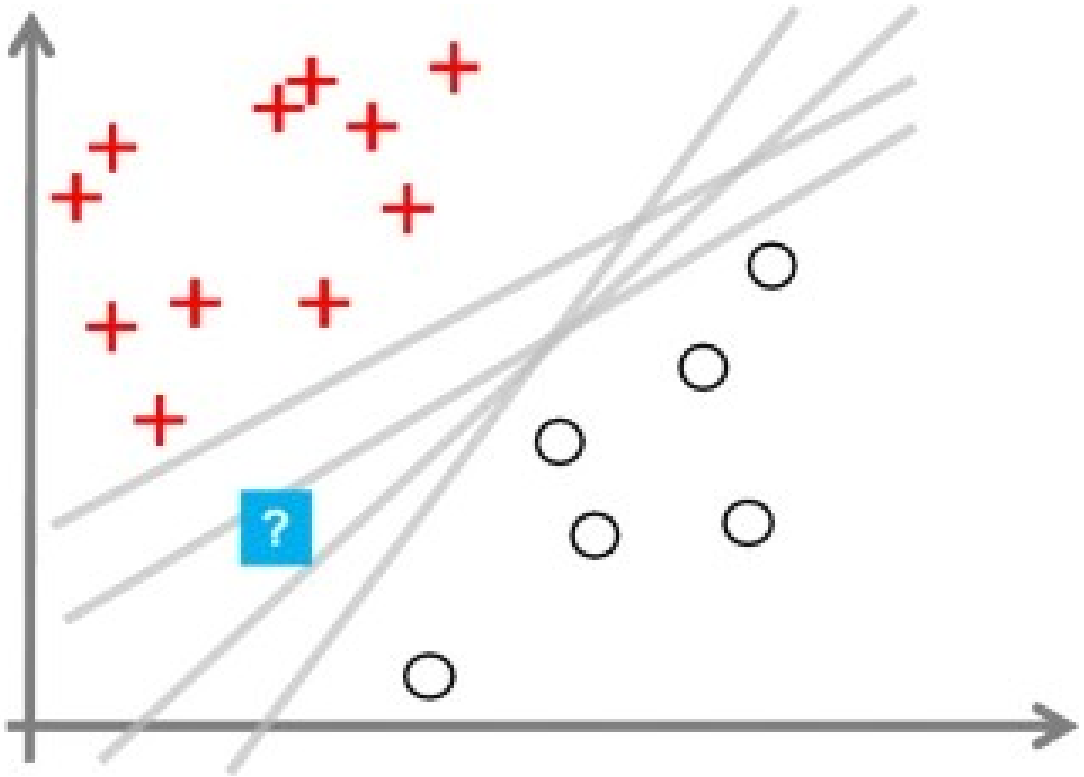
Google AI
Brain Team

Video:
https://slideslive.com/38935801/practical-uncertainty-estimation-outofdistribution-robustness-in-deep-learning

Two types of uncertainty



eatoric / fundamental / irreducible / data uncertainty

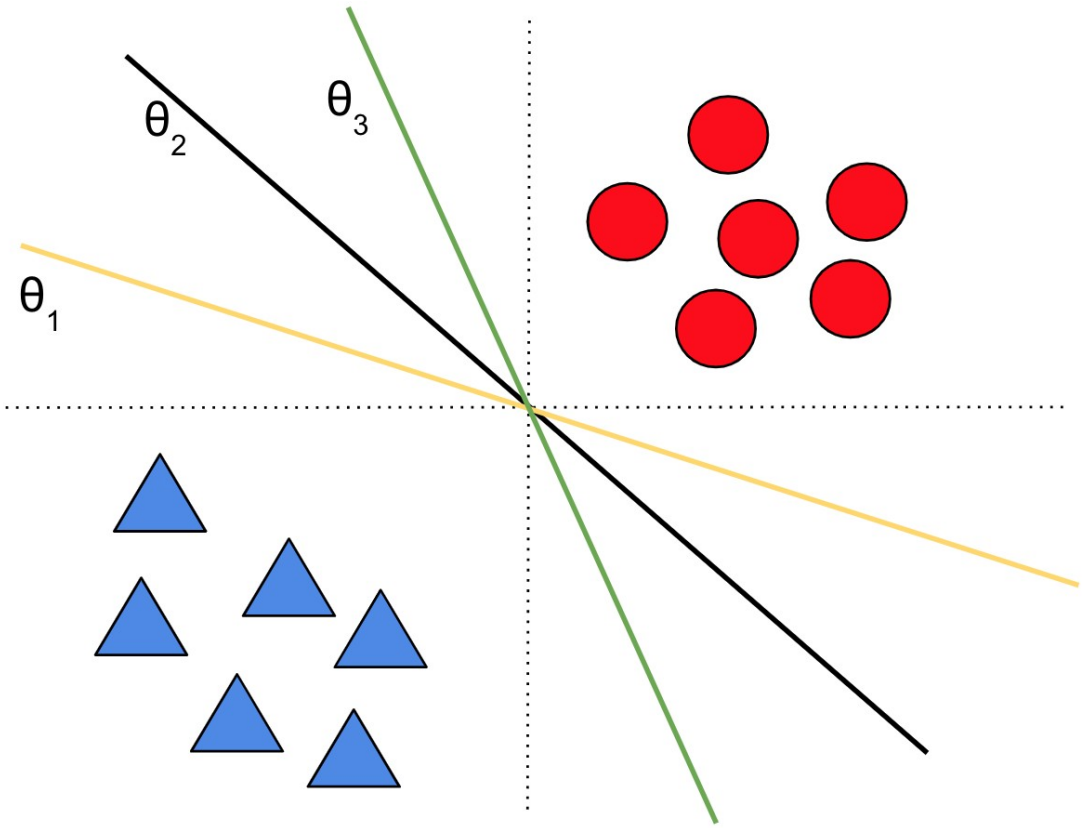


Epistemic / Model uncertainty
– due to our lack of knowledge

Do we care? Lots of applications. Right side will cause problems.

Sources of uncertainty: *Model uncertainty*

- Many models can fit the training data well
- Also known as ***epistemic uncertainty***
- Model uncertainty is "**reducible**"
  - Vanishes in the limit of infinite data (subject to model identifiability)



Sources of uncertainty: *Data uncertainty*

- Labeling noise (ex: human disagreement)

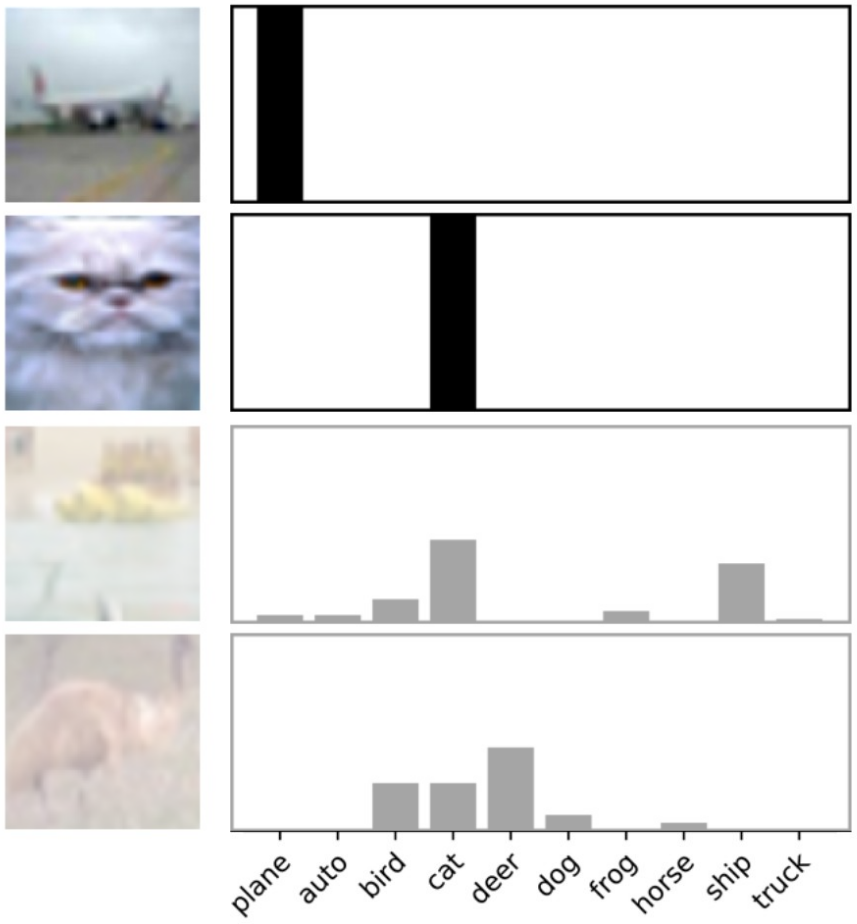


Image source: Battleday et al. 2019 “Improving machine classification using human uncertainty measurements”

Sources of uncertainty: *Data uncertainty*

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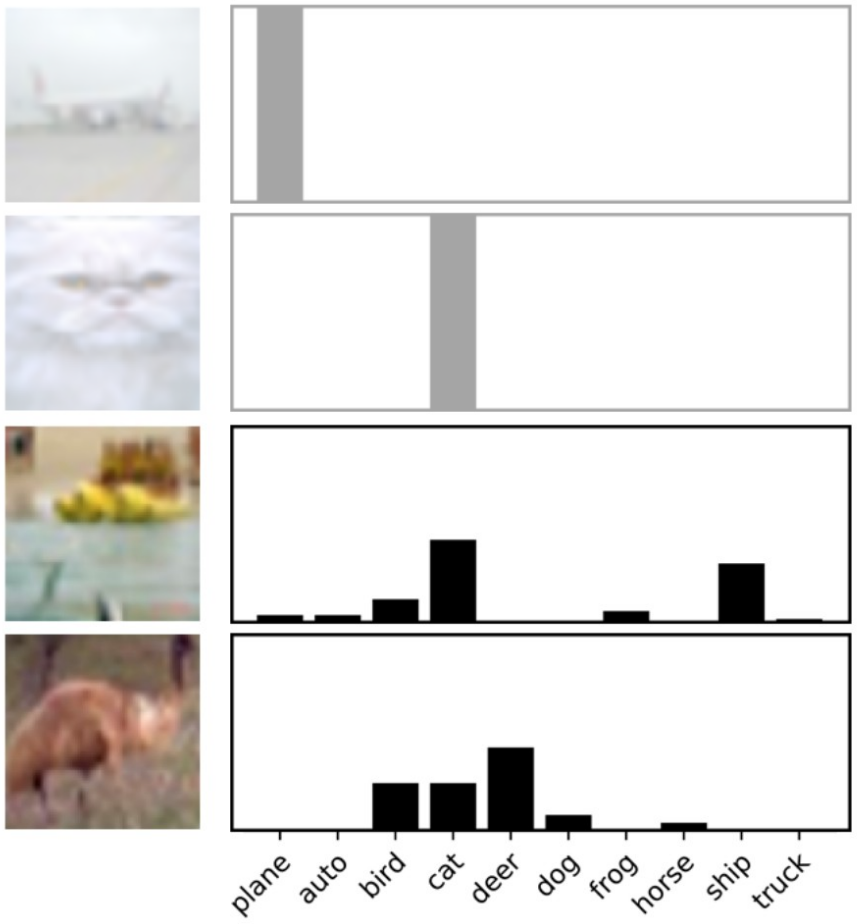


Image source: Battleday et al. 2019 “Improving machine classification using human uncertainty measurements”

Sources of uncertainty: *Data uncertainty*

- Labeling noise (ex: human disagreement)
- Measurement noise (ex: imprecise tools)
- *Missing* data (ex: partially observed features, unobserved confounders)
- Also known as ***aleatoric uncertainty***
- Data uncertainty is "**irreducible***"
  - Persists even in the limit of infinite data
  - *Could be reduced with additional

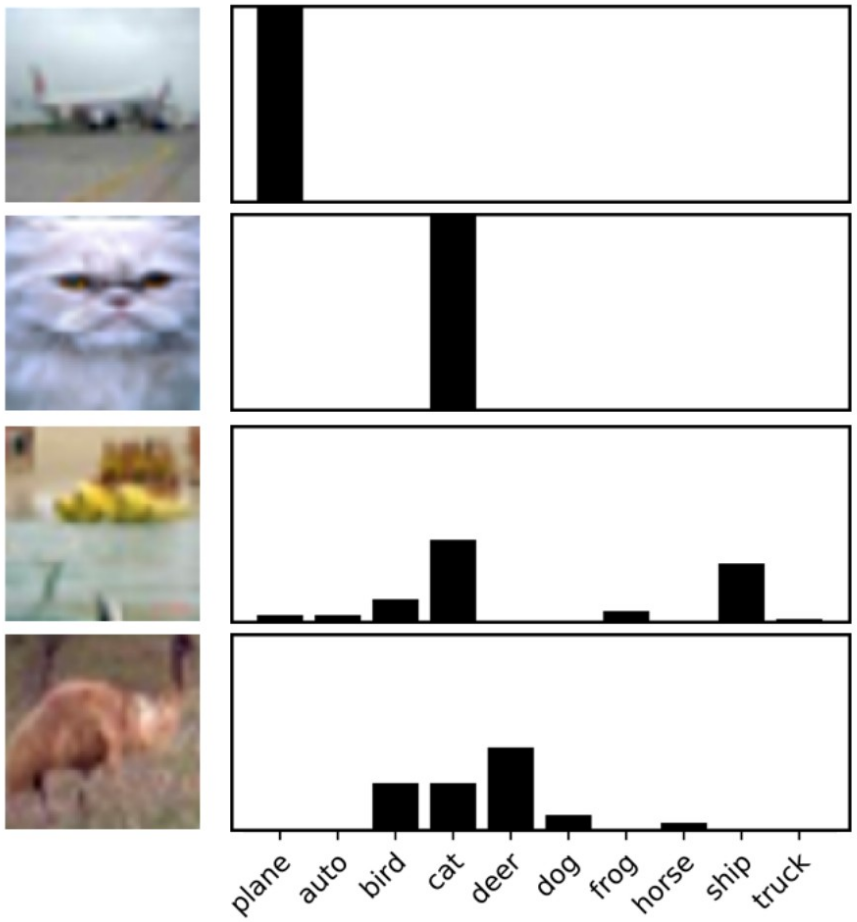


Image source: Battleday et al. 2019 "Improving machine classification using human uncertainty measurements"

How do we measure the quality of uncertainty?

$$\text{Calibration Error} = |\text{Confidence} - \text{Accuracy}|$$

predicted probability of correctness

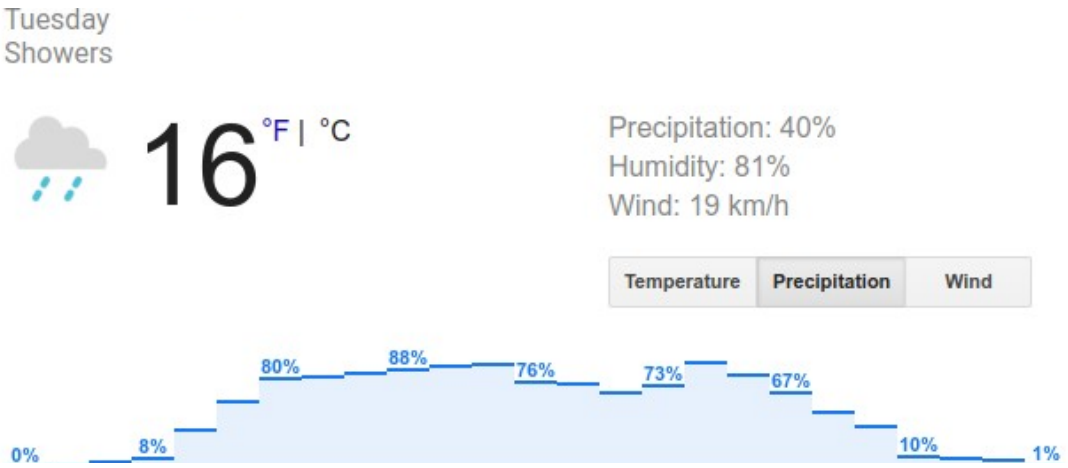
observed frequency of correctness

How do we measure the quality of uncertainty?

Calibration Error = |Confidence - Accuracy|

Of all the days where the model predicted rain with 80% probability, what fraction did we observe rain?

- 80% implies perfect calibration
- Less than 80% implies model is overconfident
- Greater than 80% implies model is under-confident



How do we measure the quality of uncertainty?

Expected Calibration Error [Naeini+ 2015]:

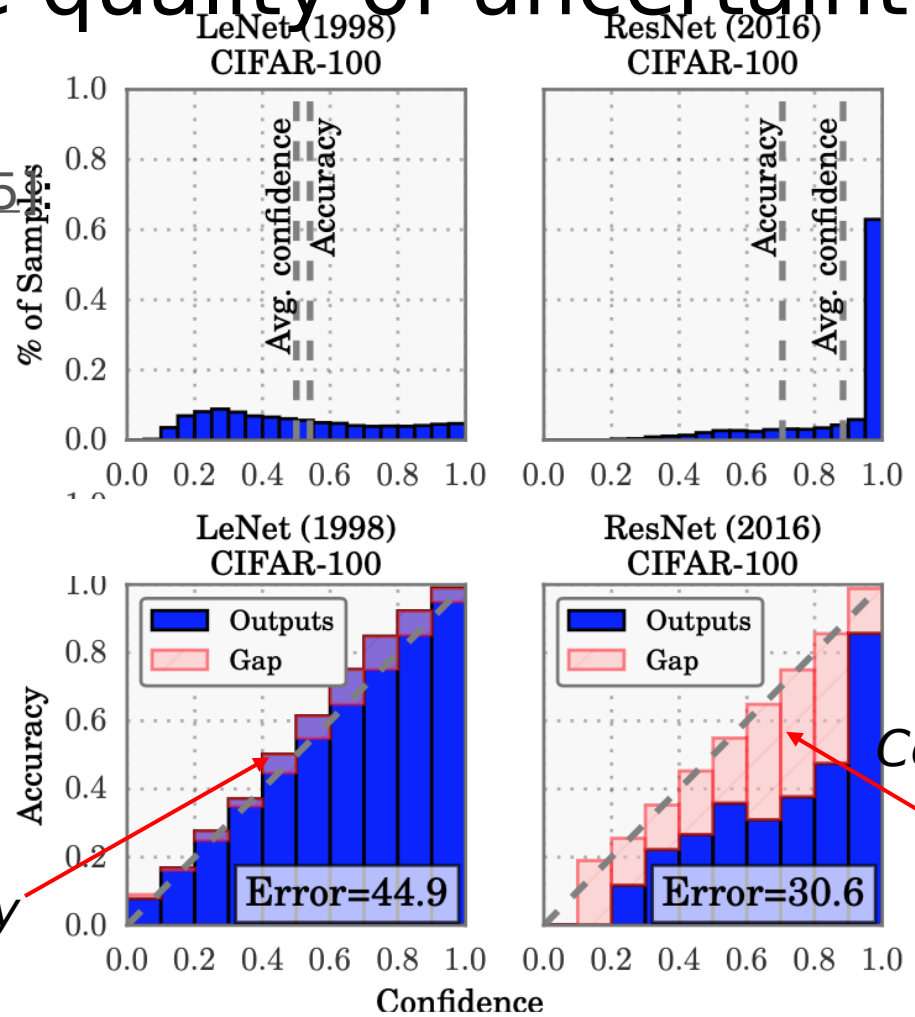
$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\mathrm{acc}(b) - \mathrm{conf}(b)\right|$$

- Bin the probabilities into B bins.
- Compute the within-bin accuracy and within-bin predicted confidence.
- Average the calibration error across bins (weighted by number of points in each bin).

How do we measure the quality of uncertainty?

Expected Calibration Error [Naeini+ 2015]:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\mathrm{acc}(b) - \mathrm{conf}(b)\right|$$



Confidence > Accuracy

=> Overconfident

Confidence < Accuracy

=> Underconfident

Image source: Guo+ 2017 “On calibration of modern neural networks”

How do we measure the quality of uncertainty?

Expected Calibration Error [Naeini+ 2015]:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\mathrm{acc}(b) - \mathrm{conf}(b)\right|$$

Note: Does **not** reflect **accuracy**.

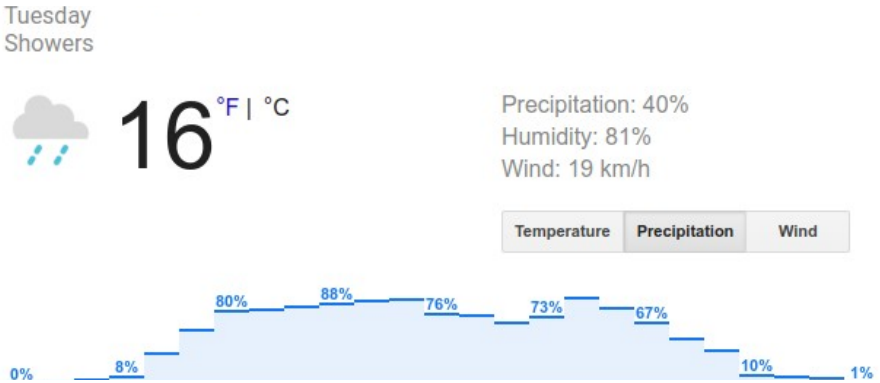
Predicting class frequency p(y=1) = 0.3 for all the inputs achieves perfect calibration.

True label	0	0	0	0	0	0	0	1	1	1	Accurate ?	Calibrated ?
Model prediction	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	❌	✅

How do we measure the quality of uncertainty?

Proper scoring rules [Gneiting & Raftery 2007]

- Negative Log-Likelihood (NLL)
  - Also known as *cross-entropy*
  - Can overemphasize tail probabilities



- Brier Score
  - Quadratic penalty (bounded range [0,1] unlike log).

$$\mathrm{BS} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \left[p(y|\mathbf{x}_n, \theta) - \delta(y - y_n) \right]^2$$

  - Can be numerically unstable to optimize.

Simple Baselines

Simple Baseline: Recalibration

For classification, modify softmax probabilities **post-hoc.**

Temperature Scaling.

Parameterize output layer with scalar T.

$$p(y_i|x) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

Minimize loss with respect to T on a separate "recalibration" dataset.

Caveat: Dataset shift...

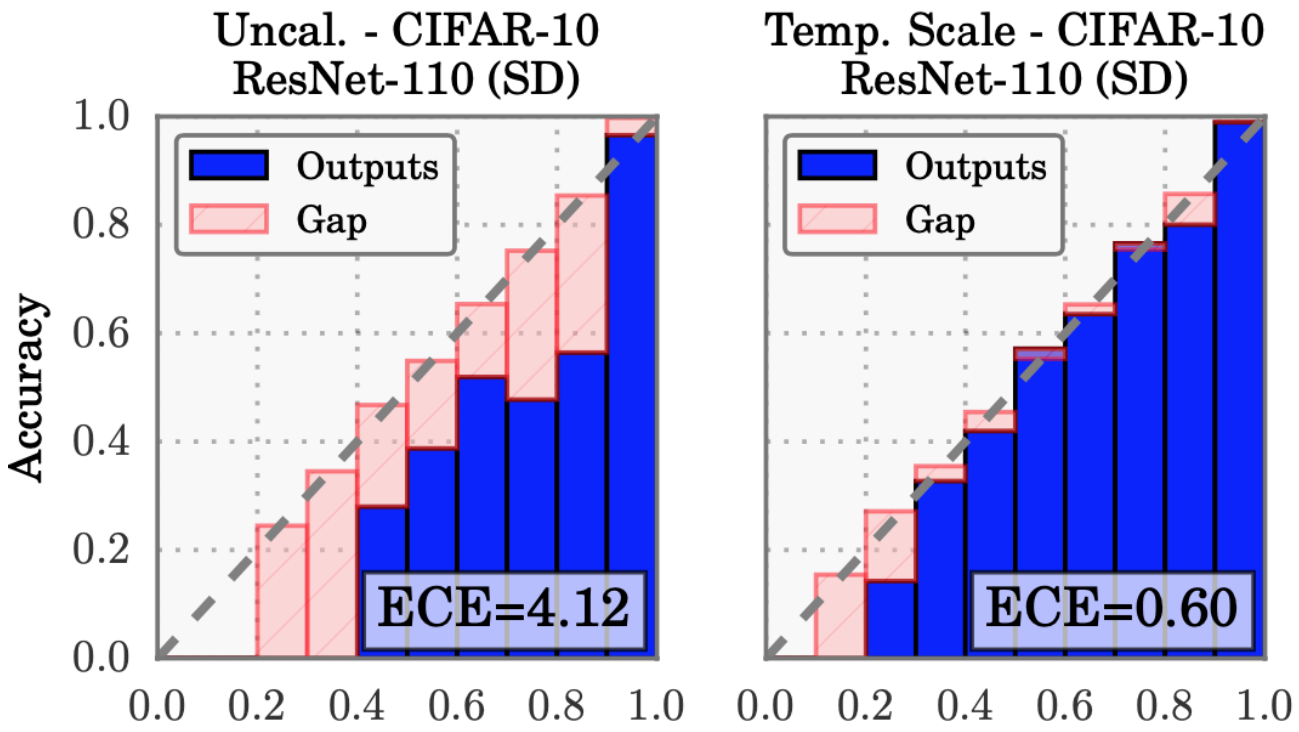


Image source: Guo+ 2017 "On calibration of modern neural networks"

Simple Baseline: Monte Carlo Dropout

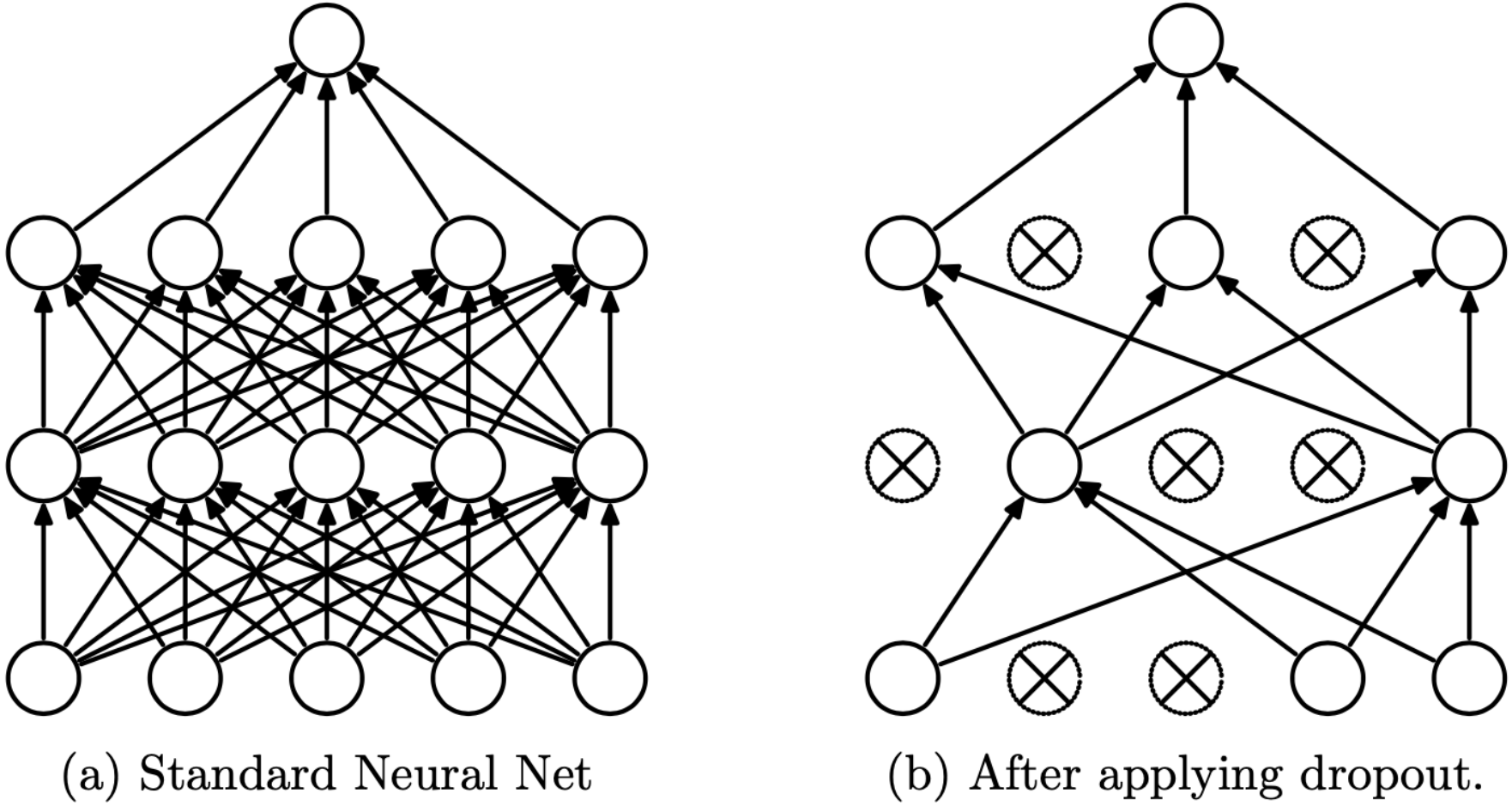


Image source: Dropout: A Simple Way to Prevent Neural Networks from Overfitting

[Gal+ 2015]

Monte Carlo Dropout - average the probabilities

- Turn on dropout…

```
model.eval()
def enable_dropout(m):
    if isinstance(m, torch.nn.Dropout):
        m.train()
model.apply(enable_dropout)
```

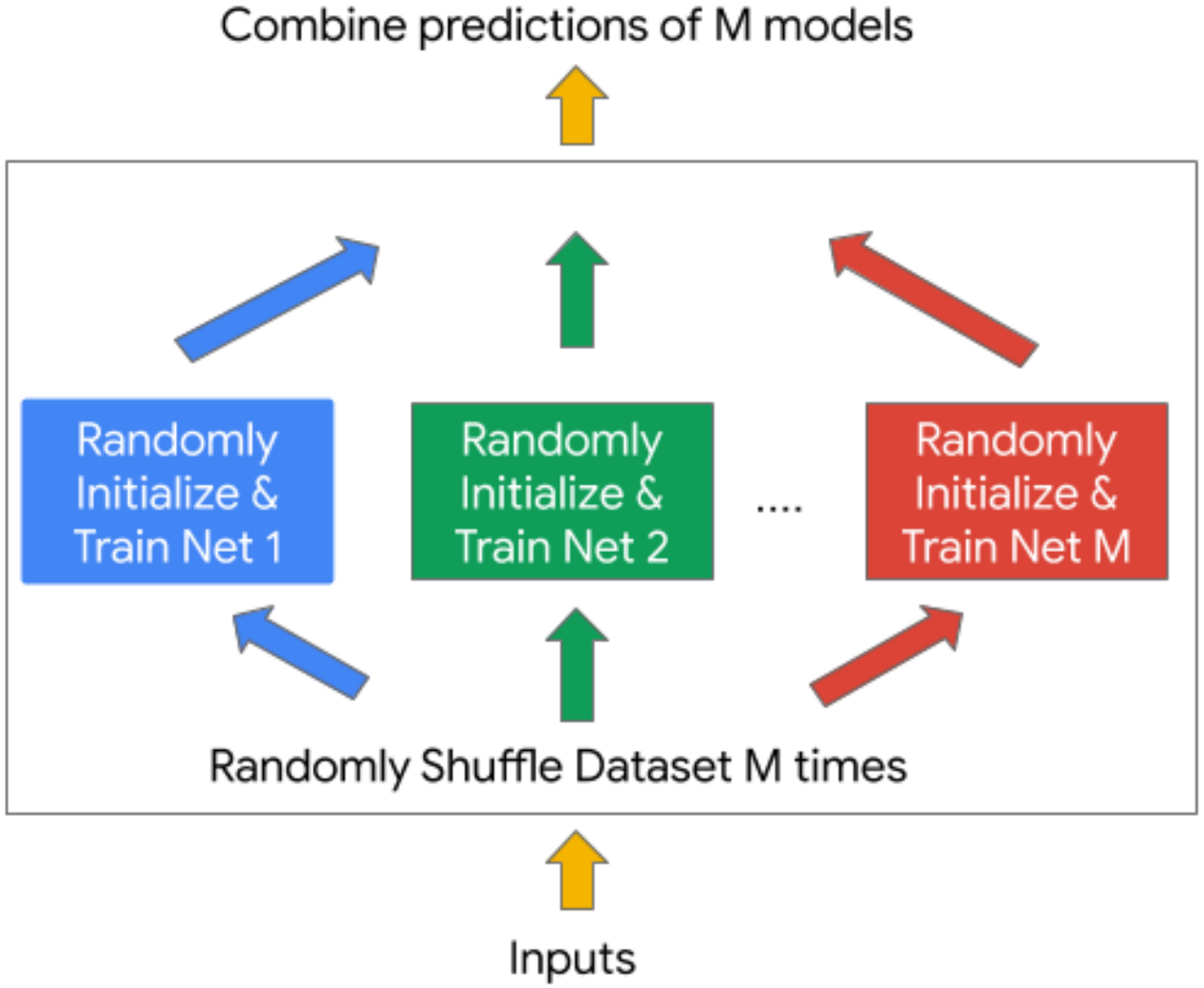
Average the probabilities (pseudo-code):

```
avg_p = 0
for i in range(n_dropout):
        logits = model(x)
        p = softmax(logits)
        avg_p += p / n_dropout
```

Simple Baseline: Deep Ensembles

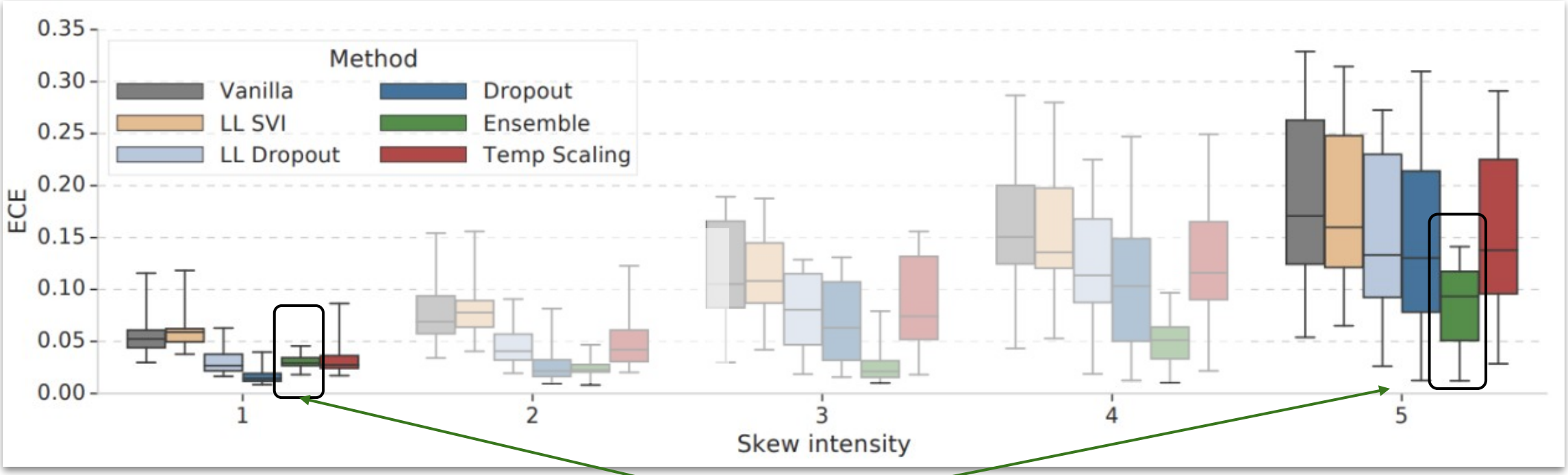
Idea: Just re-run standard SGD training but with different random seeds and average the predictions

- A well known trick for getting better accuracy and Kaggle scores



[
Lakshminarayanan+ 20

Deep Ensembles work surprisingly well in practice



Deep Ensembles are consistently among the best performing methods, especially under dataset shift

Hyperparameter Ensembles

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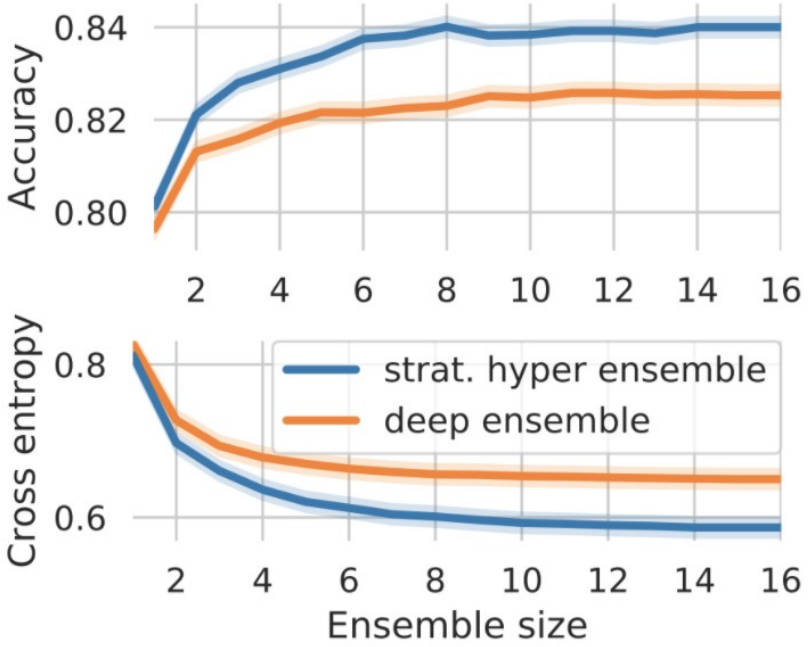
Deep ensembles differ only in random seed. By expanding the space of hyperparameters we average over, we can get even better accuracy & uncertainty estimates.

1. Run random search to generate a set of models.
   a. Include random seed as part of the search space.

2. Greedily select the K models to pool.



[Wenzel+ 2020 @ NeurIPS]

Bayesian neural networks

- We’ll get to these later in the class

Training to avoid over-confidence – label smoothing

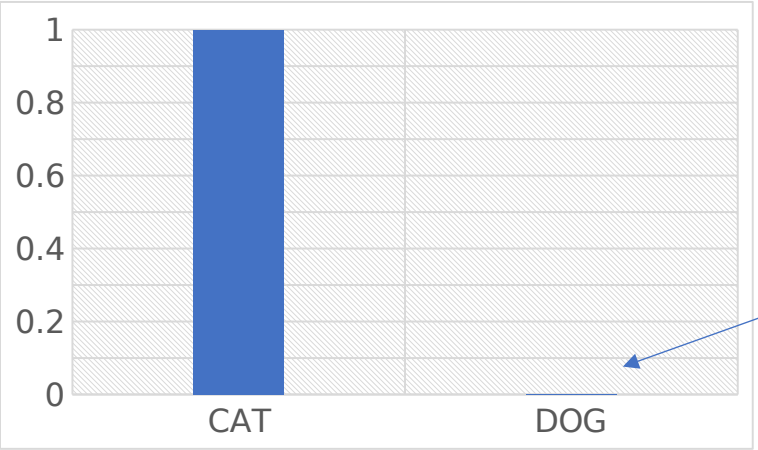
y = cat

x =

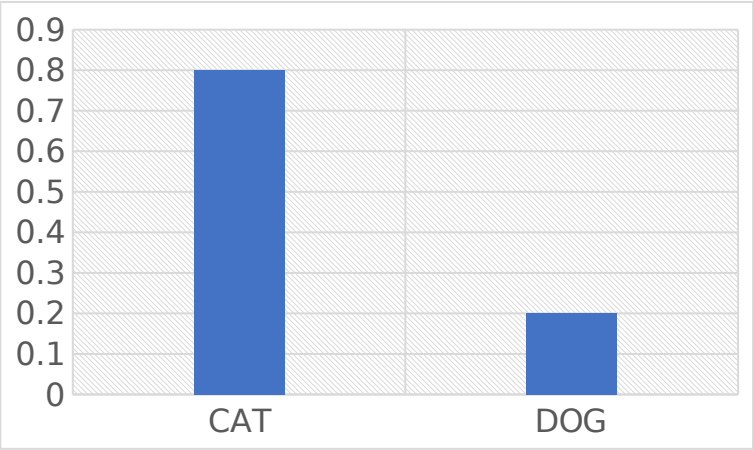
Show on board using largest marker possible

Super easy to try:

criterion = torch.nn.CrossEntropyLoss(**label_smoothing=0.1**)



Label could be wrong?



$P_{data}(y|x)$?

$P_{\phi}(y|x)$

Applications of uncertainty quantification and domain shift

Known unknowns and unknown unknowns

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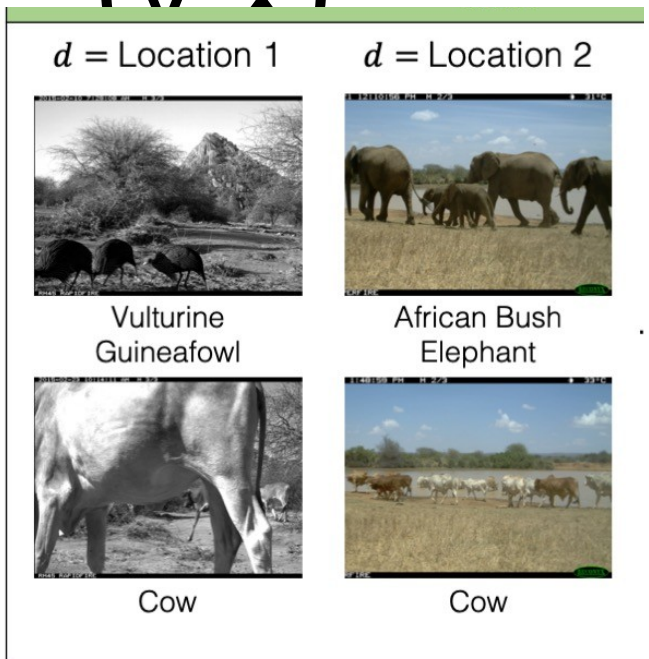
I.I.D. $p_{TEST}(y,x) = p_{TRAIN}(y,x)$

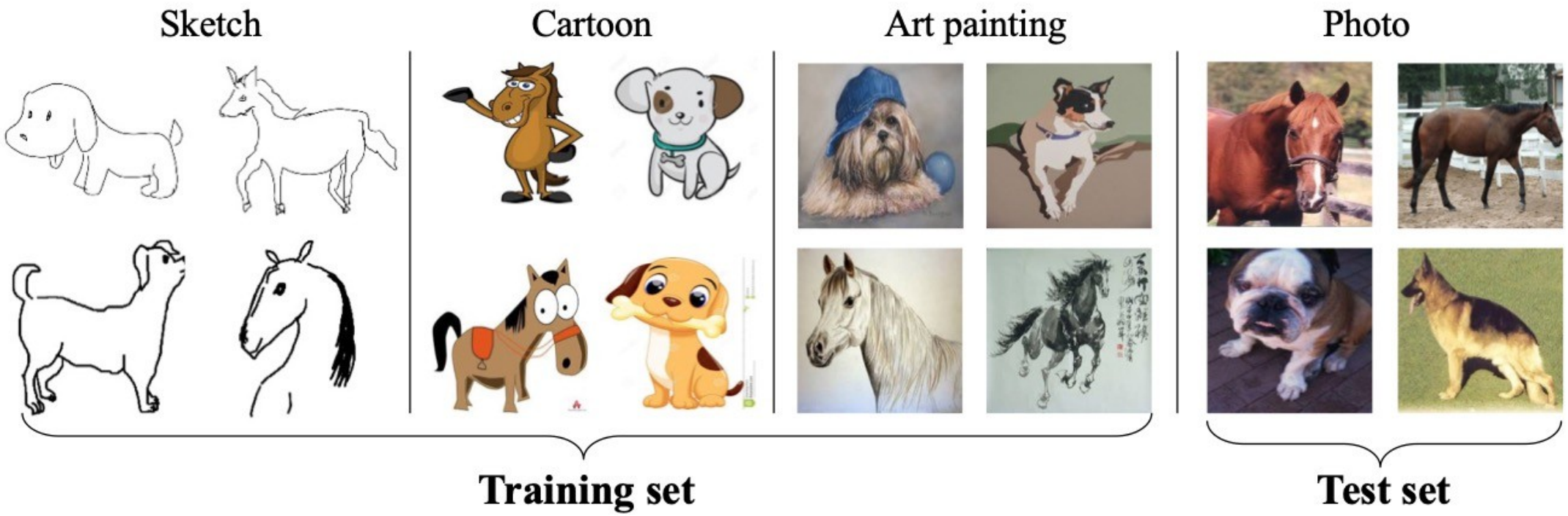
O.O.D. $p_{TEST}(y,x) \neq p_{TRA}$ (y,x)

Examples of dataset shift:

- ***Covariate shift.*** Distribution of features p(x) changes and p(y|x
- ***Label shift.*** Distribution of labels p(y) changes and p(x|y) is fixe



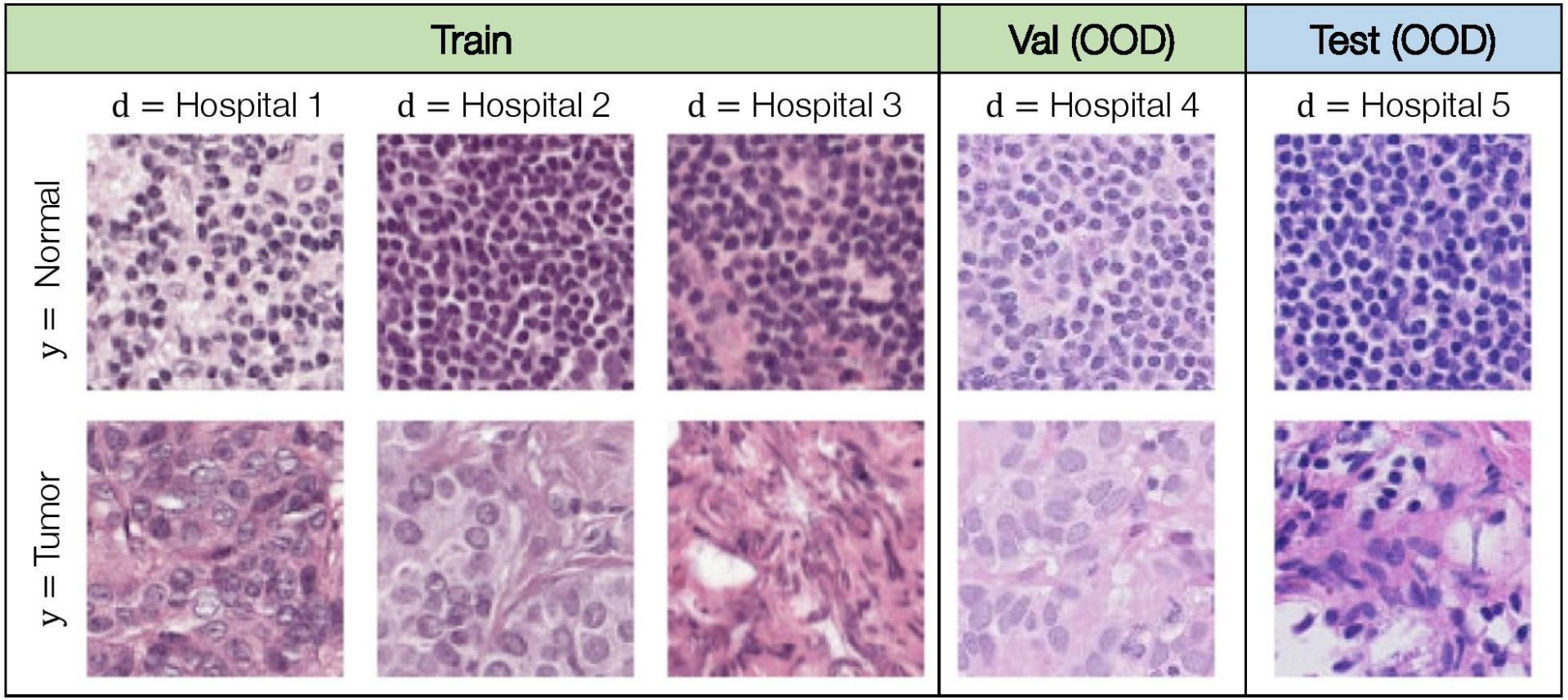
More extreme covariate shift



- $P_{Train}(x, y) \neq P_{Test}(x, y)$
- Source: Train, Target: test

Images are copied from PACS dataset [Li et al.'17]

Does the tissue contain tumor cells?



95% accuracy, useful maybe! **70% accuracy in new hosp**

Does the model *know* it is doing wor

Controllable experiments with dataset shift: ImageNet-C

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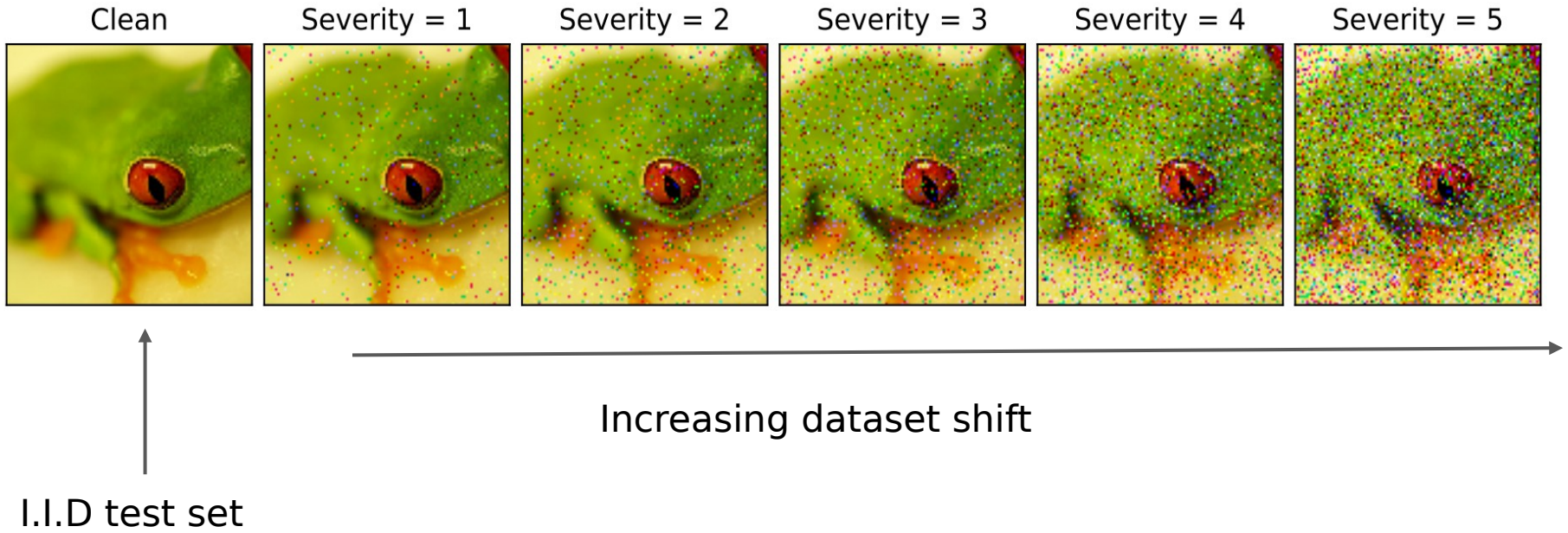

Image source: Benchmarking Neural Network Robustness to Common Corruptions and Perturbations, Hendrycks & Dietterich, 201
.

ImageNet-C: Varying Intensity for Dataset Shift

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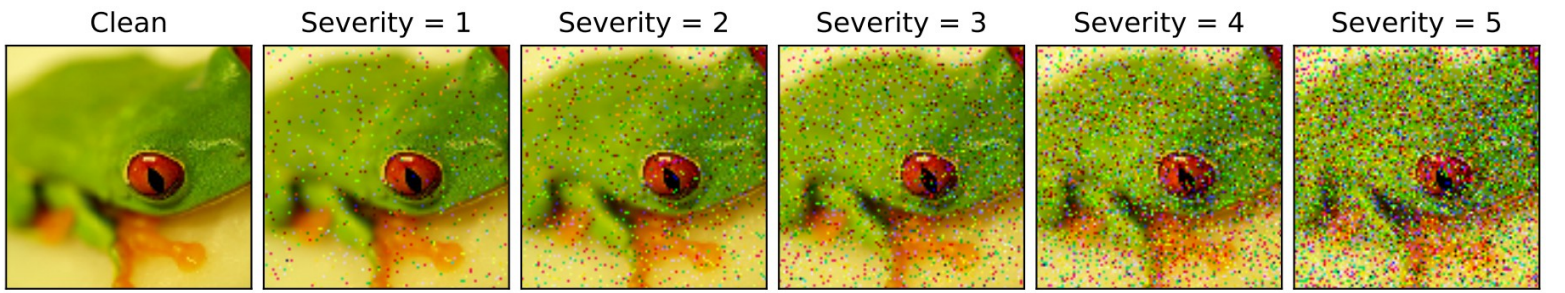

Increasing dataset shift

I.I.D test set

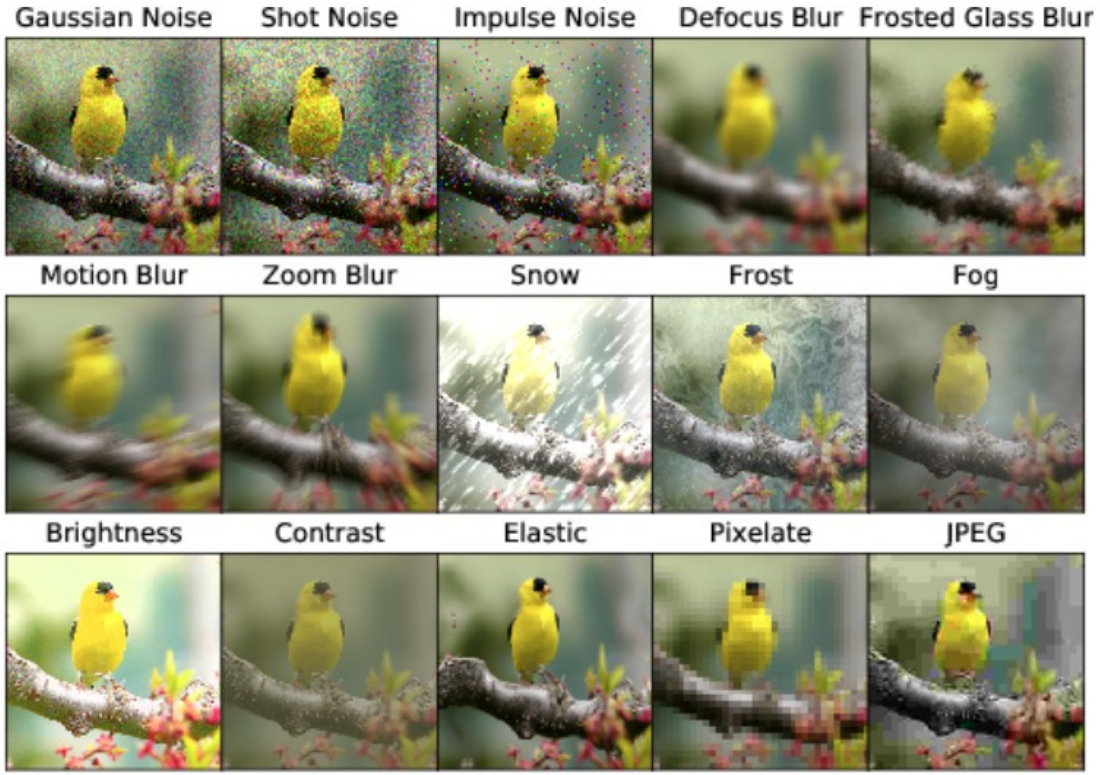

Image source: Benchmarking Neural Network Robustness to Commo
Corruptions and Perturbations, Hendrycks & Dietterich, 2019.

Neural networks do not generalize under covariate shift

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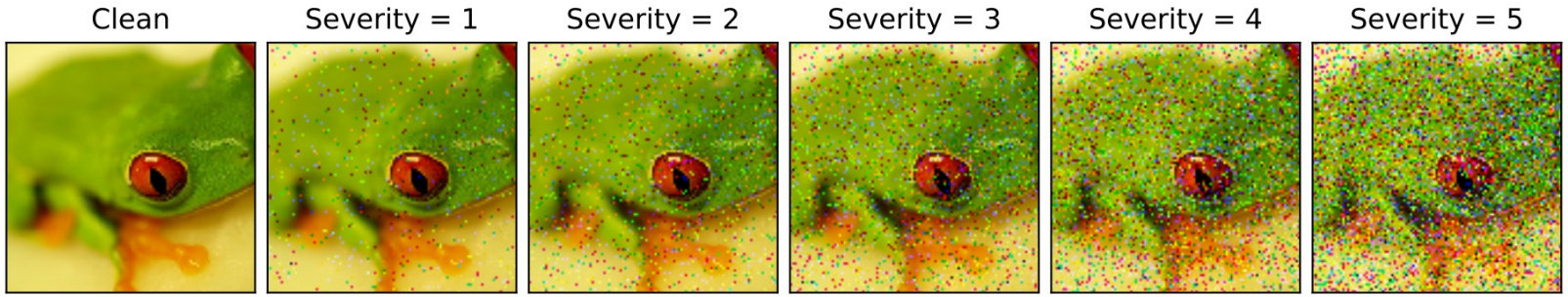

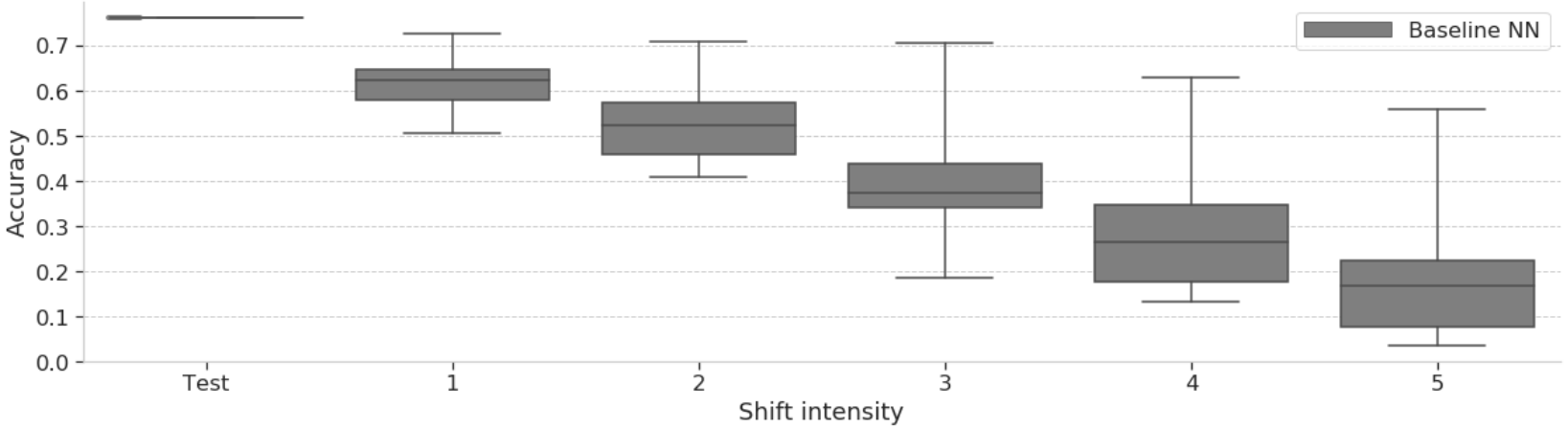

- **Accuracy drops** with increasing shift on Imagenet-C

- But do the models know that they are less accurate?

Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift?, Ovadia et al. 2019

Neural networks ***do not know when they don't know***

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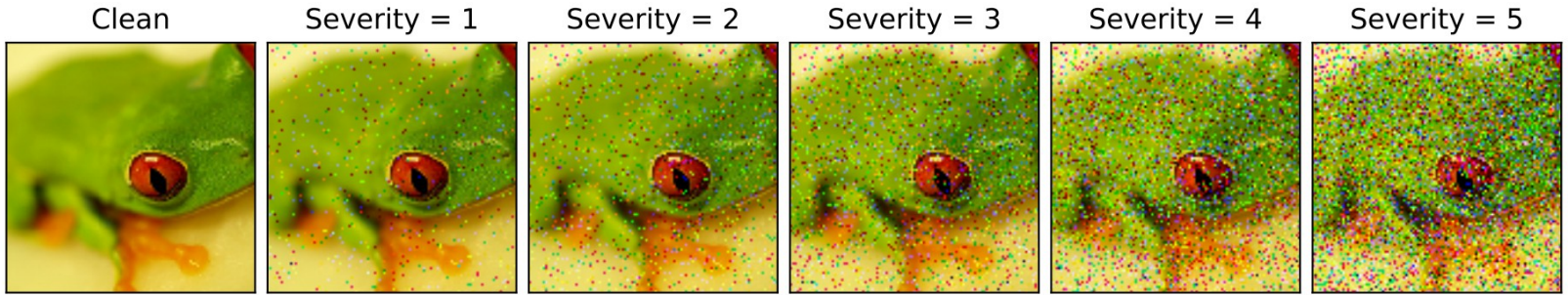

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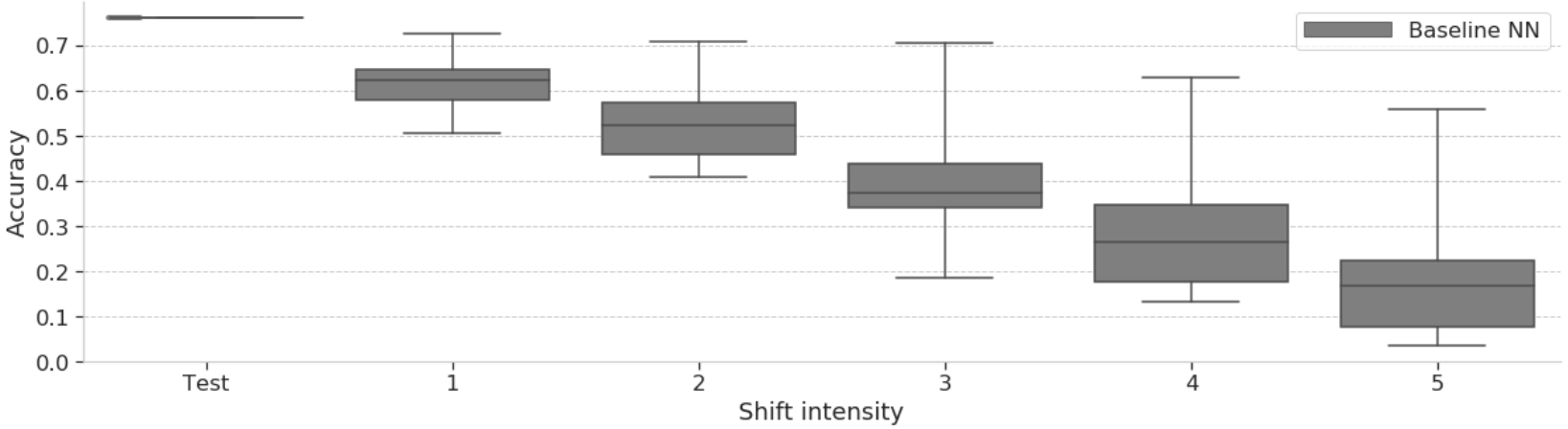
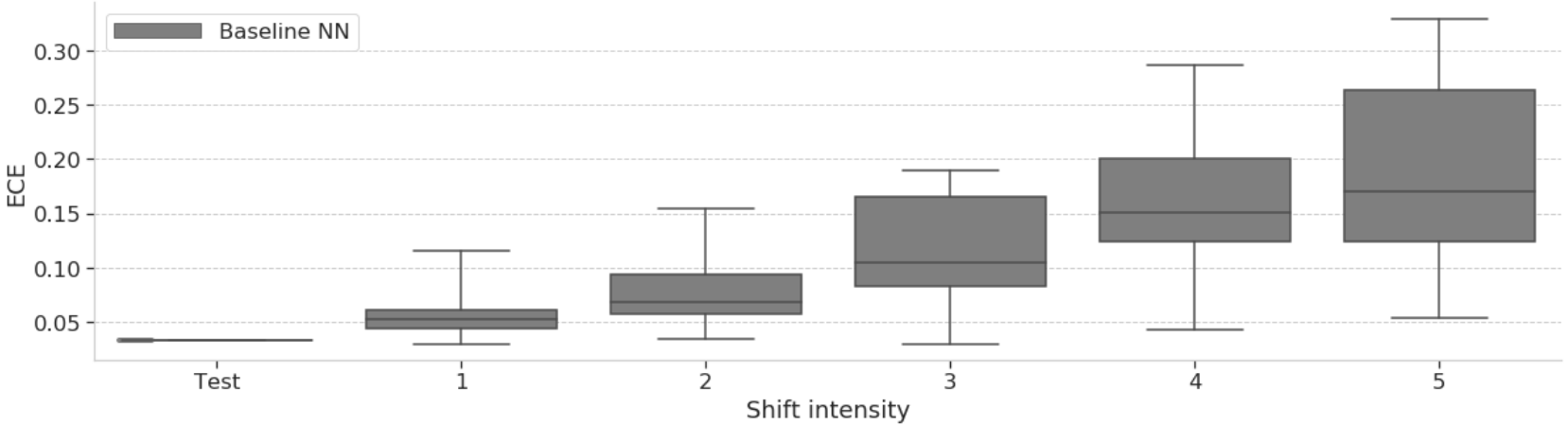


- **Quality of uncertainty degrades with shift**
  -> "overconfident mistakes"



Models assign high confidence predictions to OOD inputs

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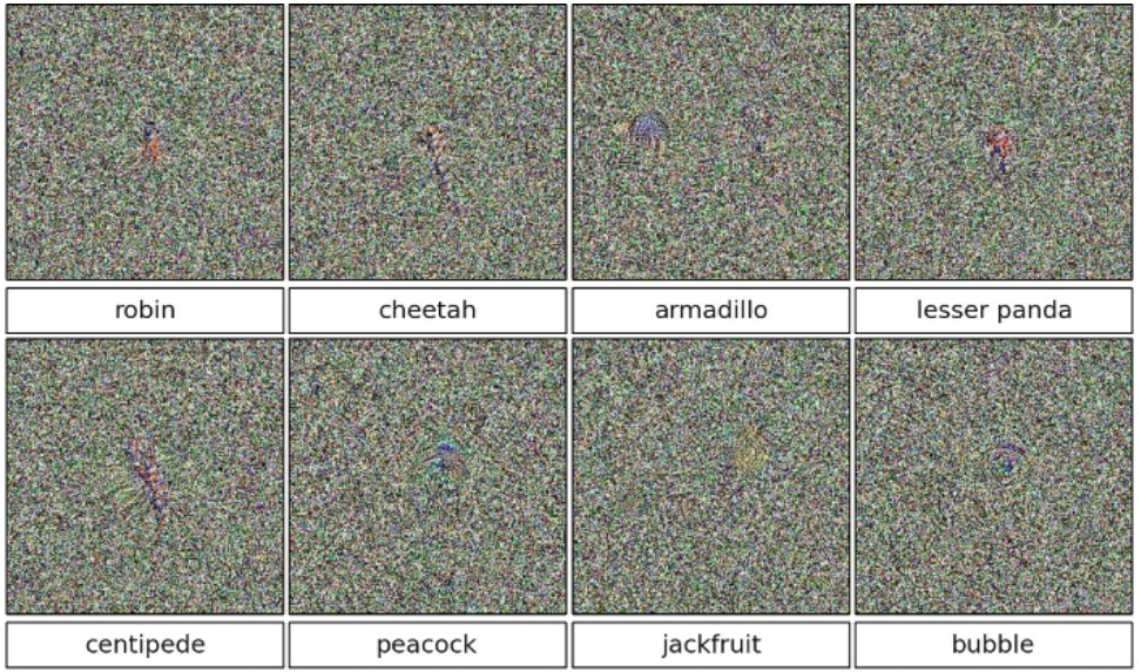
Example images where model assigns >99.5% confidence.



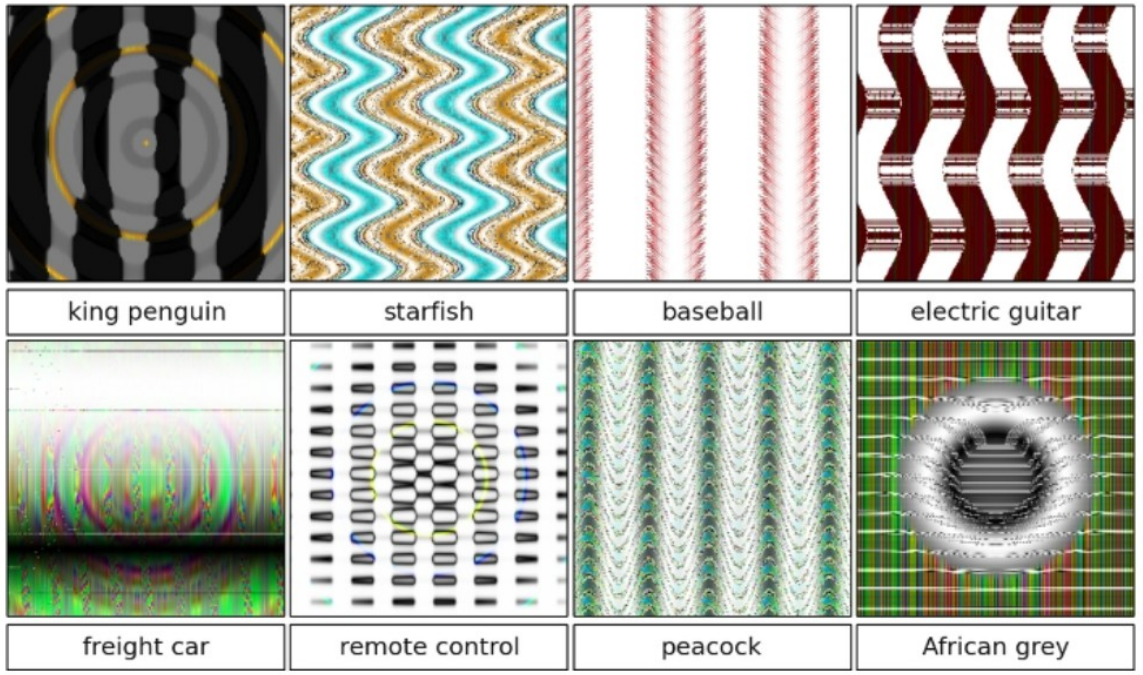
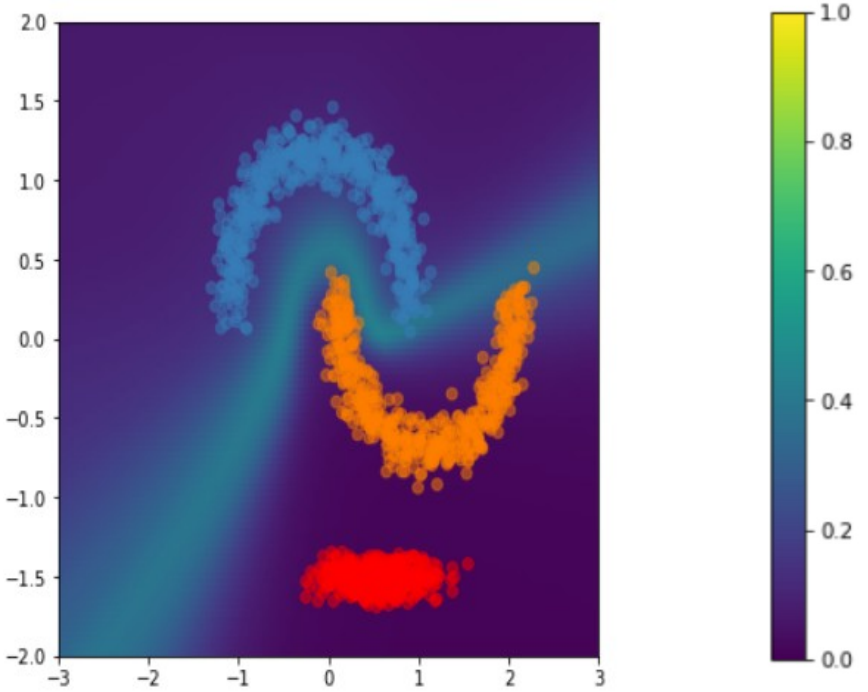


Image source: “Deep Neural Networks are Easily Fooled: High Confidence Predictions for Unrecognizable Images” Nguyen et al. 2014

Models assign high confidence predictions to OOD inputs



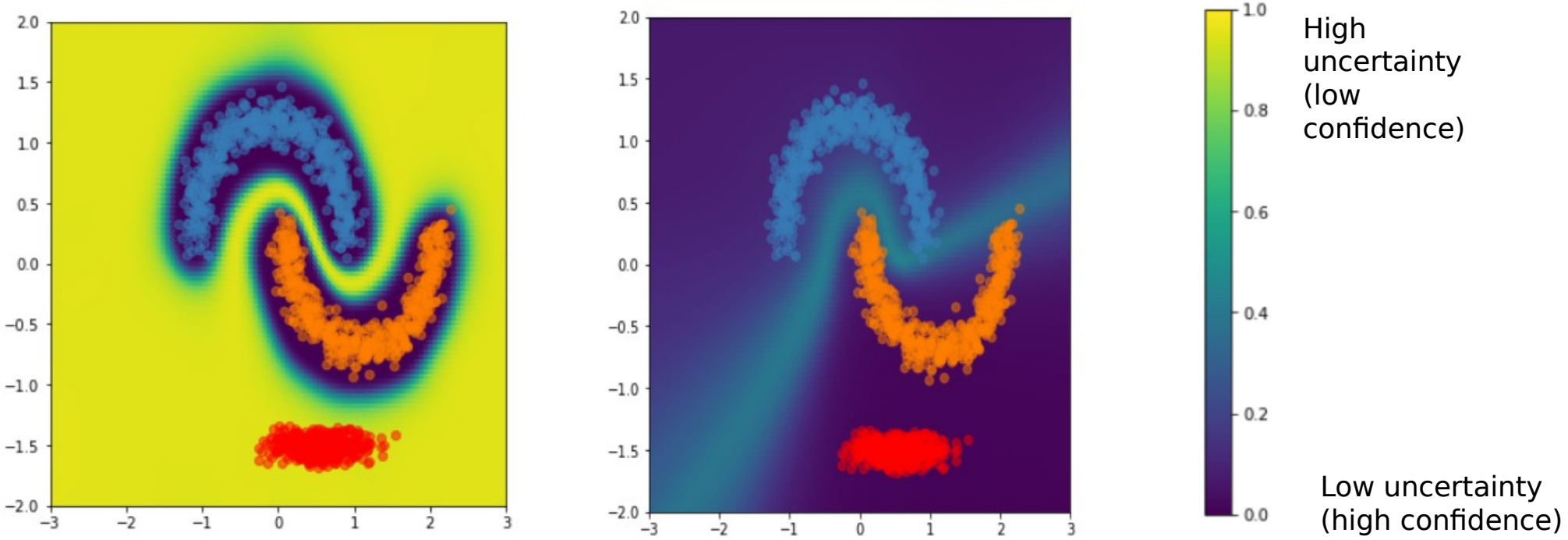
High uncertainty (low confidence)

Low uncertainty (high confidence)

Deep neural networks

Image source: “Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness” Liu et al. 2020

Models assign high confidence predictions to OOD inputs



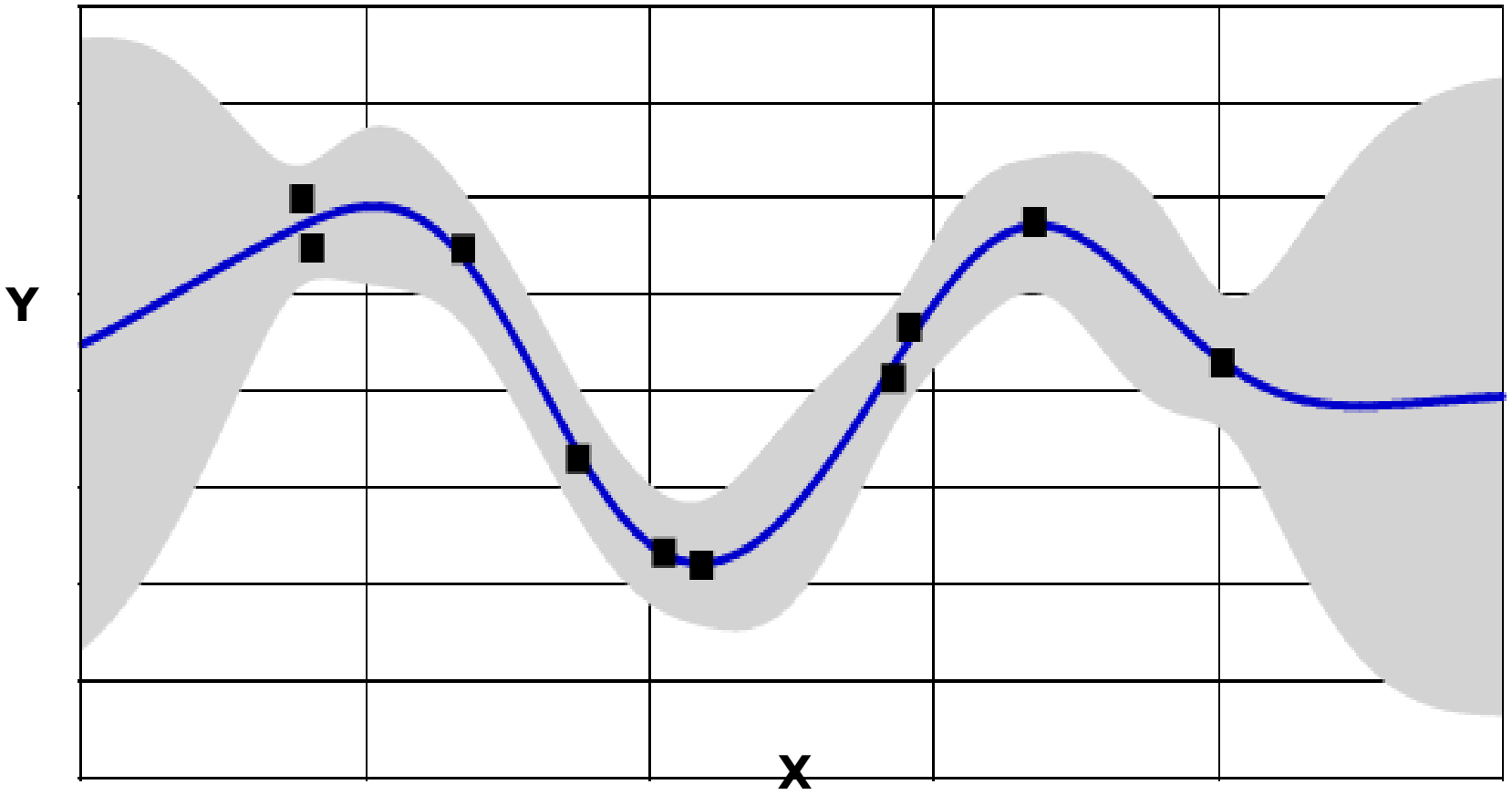
Ideal behavior

Deep neural networks

Trust model when x* is close to $p_{TRAIN}(x,y)$

Image source: "Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness" Liu et al. 2020

Aside: Uncertainty for regression



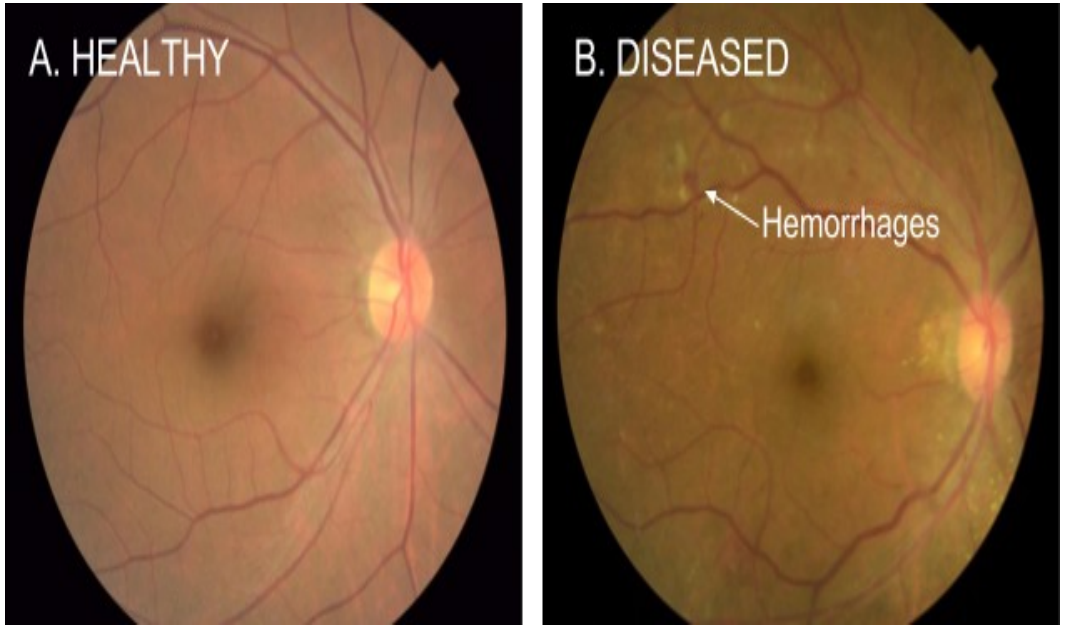
Regression (see "Conformal Prediction" literature)

Image credit: Eric Nalisnick

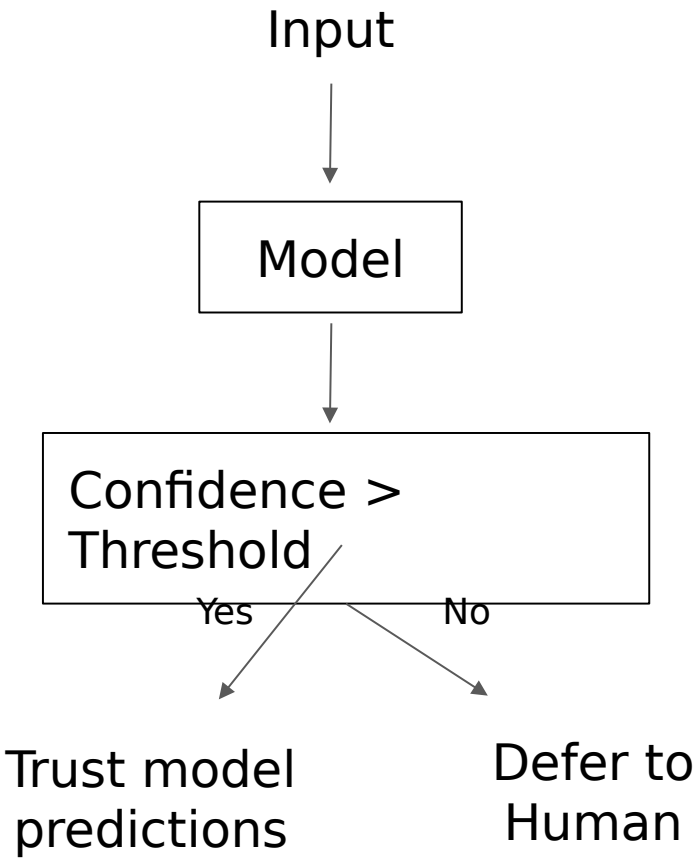
Applications

Healthcare

- Use model uncertainty to decide when to trust the model or to defer to a human.
- Reject low-quality inputs.



Diabetic retinopathy detection from fundus images [Gulshan et al, 2016]



Self-driving cars

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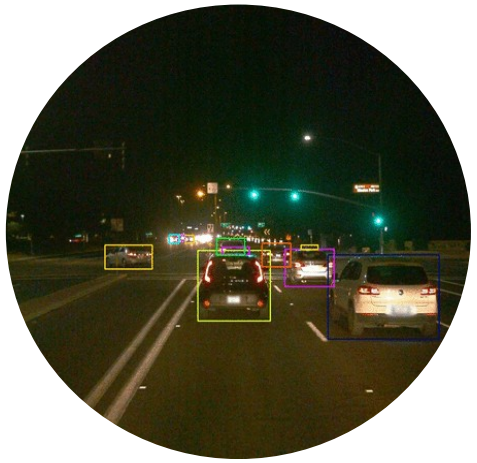
Dataset shift:

- Time of day / Lighting
- Geographical location (City vs suburban)
- Changing conditions (Weather / Construction)

Daylight



Night

Weather

Construction

Downtown

Suburban

Image credit: Sun et al, [Waymo Open Dataset]

Conversational Dialog systems

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- Detecting out-of-scope utterances

Figure 1: Example exchanges between a user (blue, right side) and a task-driven dialog system for personal finance (grey, left side). The system correctly identifies the user's query in (1), but in (2) the user's query is mis-identified as in-scope, and the system gives an unrelated response. In (3) the user's query is correctly identified as out-of-scope and the system gives a fall-back response.

Image source: Larson et al. 2019 "An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction"

Wrapping Up

Open Challenge: Scale

Enable uncertainty & robustness at the billion-trillion parameter scale.

Datasets. What is the role of priors on increasingly larger and diverse datasets?

Tasks. How do we think about OOD as we move toward general solutions to a wide range of tasks?

Model Parallelism. Mixtures of experts are already the backbone. Can we exploit recent ideas to enable even bigger and adaptive models?

Open Challenge: Benchmarks

In the past few years, there's been an ongoing call to action on benchmarks:

- *Comprehensive baselines* across standard and SOTA methods.
- *Large-scale models & datasets.*
- *High-quality code*: small changes in the setup can dramatically affect performance.

Preliminary efforts exist but a unified effort is required.

Today, we’re happy to announce we made progress on this challenge.

Uncertainty Baselines

github.com/google/uncertainty-baselines

High-quality implementations of baselines on a variety of tasks.

Ready for use: 7 settings, including:

- Wide ResNet 28-10 on CIFAR
- ResNet-50 and EfficientNet on ImageNet
- BERT on Clinc Intent Detection

14 different baseline methods.

Used across **10** projects at Google.

Collaboration with OATML @ Oxford, unifying github.com/oatml/bdl-benchmarks.

dustinvtran and edward-bot Retune VI baseline for CIFAR. ... Latest commit 9379550 3 hours ago

..

README.md	Retune VI baseline for CIFAR.	3 hours ago
batchensemble.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
batchensemble_model.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
batchensemble_model_test.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
deterministic.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
deterministic_test.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
dropout.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
dropout_test.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
ensemble.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
utils.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago
variational_inference.py	Retune VI baseline for CIFAR.	3 hours ago
variational_inference_test.py	Move baselines/cifar10/ to baselines/cifar/.	13 days ago

README.md

Wide ResNet 28-10 on CIFAR

CIFAR-10

Method	Train/Test NLL	Train/Test Accuracy	Train/Test Cal. Error	cNLL/cA/cCE	Train Runtime (hours)	# Parameters
Deterministic	1e-3 / 0.159	99.9% / 96.0%	1e-3 / 0.0231	1.29 / 69.8% / 0.173	1.2 (8 TPUv2 cores)	36.5M
BatchEnsemble (size=4)	0.08 / 0.143	99.9% / 96.2%	5e-5 / 0.0206	1.24 / 69.4% / 0.143	5.4 (8 TPUv2 cores)	36.6M
Dropout	2e-3 / 0.160	99.9% / 95.9%	2e-3 / 0.0241	1.35 / 67.8% / 0.178	1.2 (8 TPUv2 cores)	36.5M
Ensemble (size=4)	2e-3 / 0.114	99.9% / 96.6%	-	-	1.2 (32 TPUv2 cores)	146M
Variational inference	1e-3 / 0.211	99.9% / 94.7%	1e-3 / 0.029	1.46 / 71.3% / 0.181	5.5 (8 TPUv2 cores)	73M

Robustness Metrics

github.com/google-research/robustness_metrics

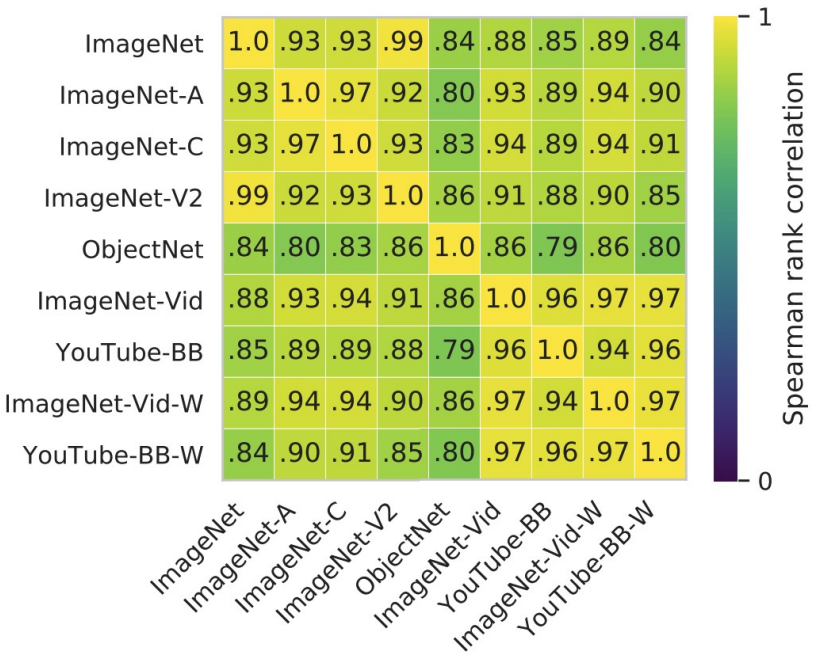
Lightweight modules to evaluate a model's robustness and uncertainty predictions.

Ready for use:

- 10 OOD datasets
- Accuracy, uncertainty, and stability metrics
- Many SOTA models (TFHub support!)
- Multiple frameworks (JAX support!)

Enables large-scale studies of robustness [Djolonga+ 2020].

Collaboration lead by Google Research, Brain Team @ Zurich.



Takeaways

- Uncertainty & robustness are critical problems in AI and machine learning.
- Benchmark models with calibration error and a large collection of OOD shifts.
- Probabilistic ML, ensemble learning, and optimization provide a foundation.
- The best methods advance two dimensions: combining multiple neural network predictions; and imposing priors and inductive biases.
- As compute increases, the uncertainty-robustness frontier outlines future progress.

My takeaways...

”Sophisticated” ML researchers think about uncertainty

There’s no obvious universal approach

More success / hype on improving accuracy, compared to uncertainty estimates, but maybe that will change as GPT models saturate at human performance